

Sap flow low-cost sensors integrated in machine learning and collaborative IoT ecosystems for the smart irrigation of olive: testing and calibration with gravimetric data

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Abstract

Effective water management in agriculture demands precise irrigation control, yet high costs and technical challenges often restrict farmers' access to advanced tools. This study presents the testing and calibration of low-cost, sap flow sensors (SFS) included in the automatable system SOFIS (Smart-Orchard Ferti-Irrigation System), an affordable, modular, and customizable digital solution for real-time irrigation scheduling, which combines sensors, IoT and machine learning to analyze in real time soil, plant, and atmospheric conditions. SFS are based in the heat pulse methodology but received extensive improvements and simplifications to provide a simple, inexpensive and precise device. The main purpose of the present study is to provide reliable estimates of plant transpiration measurement and validate the ability of the SFS integrated in the SOFIS system to provide reliable estimates of olive transpiration, performing SFS calibration and testing. Calibration of SFS installed in olive trees was performed under controlled conditions in a greenhouse, during the summer/fall of 2025, using a set of load cell systems, working as precision weighing lysimeters, to provide soil water storage variation. The preliminary results indicate an underestimation of transpiration by SFS below 10% for daily values. Actual mean daily transpiration was close to 2 and 0.5 L/d per plant for the periods of July and October, respectively. Further developments will focus on investigating whether the SFS calibration parameters are transferable within the 'Cobrançosa' olive variety to other olive varieties and whether the magnitude of the error is stable.

Keywords: 'Cobrançosa', crop monitoring crop water requirements, plant transpiration, water balance,

INTRODUCTION

The measurement of sap flow (SF) using thermal methods relies on analyzing the temperature field generated around a transient heat source. This thermal response can be described by the one-dimensional advection–diffusion equation, whose Green's function defines the spatial and temporal evolution of temperature following a heat pulse (Marshall, 1958). Under no-flow conditions, the temperature field remains symmetric around the heater; however, when SF occurs, advection displaces the heat plume downstream, creating a measurable thermal asymmetry.

For small advective heat velocities (V_h , m s⁻¹), the solution can be linearized, resulting in a first-order perturbation of the purely diffusive solution. In this framework, variations in the maximum temperature rise ($\Delta T_{\max}$, °C) are directly proportional to SF, providing a physical link between heat transport and sap movement. Nevertheless, parameters such as

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the thermal diffusivity (D , $\text{m}^2 \text{s}^{-1}$), volumetric heat capacity (ρc , $\text{J m}^{-3} \text{K}^{-1}$), and sensor–wood contact vary in time, introducing uncertainty. Thus, a hybrid approach was developed, combining the theoretical advection–diffusion framework with an empirical scaling based on evaporative demand. Daytime SF is assumed to scale linearly with vapor pressure deficit (VPD, kPa). The advective velocity can be expressed as Equation 1:

$$Vh(t) = \Lambda \frac{VPD(t)}{\Delta T(t)/\Delta T_0} \quad \text{Eq. 1.}$$

where Λ ($\text{m} \cdot \text{s}^{-1} \cdot \text{kPa}^{-1} \cdot ^\circ\text{C}^{-1}$) is an empirical coefficient that incorporates the effects of D , sensor geometry, and unit conversions and the baseline thermal response (ΔT_0 , $^\circ\text{C}$).

This study reports on the testing and calibration of low-cost, SF sensors (SFS) included in the system SOFIS® (Orchard Ferti-Irrigation System, Siqueira et al., 2022), designed to be affordable, modular, and customizable, a digital solution for real-time irrigation scheduling. SOFIS combines sensors, IoT and machine learning. The incorporated SFS, although based in the single probe heat pulse methodology (López-Bernal et al., 2017), received improvements and simplifications, such as the adoption of the Biot-Granier sensor configuration (Siqueira et al., 2020), as it enables the acquisition of absolute temperature values through a probe with the sensing element positioned at the needle tip and allows flexible operation. The main aim is to make SFS progressively simple, inexpensive and robust allowing: i) a qualitative assessment of transpiration (T) for SOFIS operation (e.g., in relative terms), ii) T quantitative assessment after calibration, iii) an evaluation of plant water status.

The main objective of the present study was to validate the ability of the SFS to provide reliable estimates of olive T, performing SFS calibration and testing, along with developing a methodology to estimate plant water use. Calibration of SFS installed in olive trees was performed under controlled conditions, during the summer/fall of 2025, using a set of load cell systems.

MATERIALS AND METHODS

Plant material and irrigation

Experiments were conducted indoors during two periods in 2025, in July (day of year, DOY 184–194) and in October (DOY 290–296), at ISA, University of Lisbon, Portugal. Olive trees (*Olea europaea* L. ‘Arbequina’), with an average stem diameter of 3 cm, were grown in 50 L pots filled with a commercial substrate. Plants were maintained under well-watered conditions (100% of field capacity) through daily (during July) or every 5 days (during October) drip irrigation. Three plants were selected for replication, although technical problems with plant 2 sensors led to some periods with missing data. The advective heat velocities computations (V_h) for subsequent estimated sap velocities were compared against reference ranges reported in the literature for similar potted olive trees under well-watered or near-saturated conditions (commonly from few to several tens cm h^{-1} , Lopez-Bernal et al., 2010). Tree transpiration was estimated considering an average cross-sectional area of the trunk of 7 cm^2 and that all the xylem area was active for sap transport, as the plants were young.

Instrumentation

Single-probe heat-pulse sensors were constructed with the following components: heater composed by a constantan resistance wire ($\varnothing$ 0.08 mm, 150 mm in length) inserted inside a metallic tube ($\varnothing$ 2 mm, 10 mm in length), calibrated to release $4286 \text{ J} \cdot \text{m}^{-1}$ over a 60 s pulse; thermistor (NTC 100 k Ω , $\beta = 4072 \text{ K}$) encapsulated at the distal tip of the probe,

reaching a final depth of 10 mm into the sapwood. The geometry consists of a 5 mm tangential separation between the heater centre and the thermistor ($y = 5$ mm), with no radial offset ($x = 0$ mm), both positioned at the same height within the stem. The probe was inserted horizontally into the stem, near 20 cm above the soil surface, with the pot sealed to prevent soil evaporation. Soil conditions (temperature, volumetric water content and electrical conductivity) in each pot were monitored using a THC soil sensor (model CWT-Soil-THC-S).

Precision weighing lysimeters built with load cells (AP06 - Beijing True-Tec Co., Ltd), and wood platforms as described in (Paço et al., 2025), were used to measure water mass variations to quantify T by water balance. The data were collected on a CR3000 datalogger (Campbell Scientific, Inc., Utah, USA) and stored every minute.

Vapor pressure deficit computation

VPD computation was based on air temperature and relative humidity measurements using a digital sensor (SHT30, ComwinTop, China; accuracy ± 0.3 °C and ± 2 % RH, range -40 to $+125$ °C, 0 – 100 % RH), installed 10 cm below the leaf apex and synchronized with each heat-pulse measurement. This variable was selected because it is the primary atmospheric driver of transpiration and thus regulates SF in olive trees (Fernandes-Silva et al., 2025), as demonstrated by relationships between SF and VPD under well-watered conditions (Rodriguez-Dominguez et al., 2019; Tognetti et al., 2004).

Algorithm and parameter calibration

The calibration parameters Λ and ΔT_0 (Eq. 1) were recalculated daily. ΔT_0 was determined using early-morning (00:00–05:00 h) measurements under minimal SF (heat transport dominated by diffusion). Λ was recalculated for each day from the observed $\Delta T_{\max}$ and maximum VPD ($VPD_{\max}$), assuming a reference maximum V_h ($V_{h\max}$). The coefficient Λ accounts for effective thermal diffusivity (D), heater–thermistor geometry, and ρc , as well as sensor–wood contact quality and local sapwood water content. The raw thermal pulse data were processed using an automated algorithm that performs daily recalibration of the baseline thermal rise (ΔT_0) and Λ . For each day, ΔT_0 was estimated as the mean ΔT between 00:00 and 05:00 h (minimal SF and heat transport dominated by conduction).

Daily $\Delta T_{\max}$ and $VPD_{\max}$ were extracted from the same dataset. Because independent measurements of SF were not available, the empirical coefficient Λ was calibrated daily by enforcing a physically plausible reference $V_{h\max}$, set to $50 \text{ cm}\cdot\text{h}^{-1}$ based on preliminary observations for well-watered potted olive trees. Λ was computed from the daily maxima of VPD and thermal response using Equation (2).

$$\Lambda = \frac{V_{h\max}/(3.6 \times 10^4)}{VPD_{\max}(\Delta T_{\max} - \Delta T_0)} \quad \text{Eq. 2.}$$

This coefficient, along with the daily baseline ΔT_0 , was stored as a daily coefficient table, automatically generated by the algorithm for continuous velocity estimation. Instantaneous V_h was then estimated for each measurement timestamp using Equation 3.

$$V_h(t) = \Lambda VPD(t) (\Delta T(t) - \Delta T_0) \quad \text{Eq. 3.}$$

Velocity was expressed in $\text{cm}\cdot\text{h}^{-1}$ by multiplying by the conversion factor 3.6×10^4 .

All computations were performed using Python 3.12 (Visual Studio Code environment) employing the pandas and NumPy libraries for data processing and numerical analysis.

Estimation of thermal diffusivity (D)

Thermal diffusivity (D , m^2s^{-1}) was estimated using a hybrid approach (advective and purely diffusive regimes). For each measurement period, D was computed considering estimated $V_h > 0$ and D obtained from the linearized advection–diffusion relation (Eq. 4).

$$D = \frac{V_h y}{2\left(\frac{\Delta T}{\Delta T_0} - 1\right)} \quad \text{Eq. 4,}$$

where $y(\text{m})$ is the tangential distance between the heater centre and the thermistor tip (5 mm), and ΔT_0 is the baseline temperature rise under non-transpiring conditions. In contrast, during non-advective periods ($V_h \leq 0$), D was computed assuming purely diffusive heat transport, using the analytical relation for the time-to-peak of the temperature pulse (Eq. 5).

$$D = \frac{y^2}{4t_{\max}} \quad \text{Eq. 5,}$$

where $t_{\max}$ is the time required for the temperature signal to reach its maximum after the heater pulse. From observations across all probes, this peak occurred 62 s after the heater was turned off, indicating that $t_{\max} \approx 62$ s was representative of the diffusive regime. This value was adopted for nighttime periods when advection was negligible.

Estimation of volumetric heat capacity

The volumetric heat capacity (ρc , $\text{J m}^{-3} \text{K}^{-1}$) was calculated from the analytical solution of the one-dimensional heat diffusion equation for a short-duration line heat pulse (Marshall, 1958; Vandegehuchte and Steppe, 2012a). The temperature rise at the thermistor (ΔT , K) at time $t_{\max}$ (s) is given by Equation 6:

$$\Delta T(t_{\max}) = \frac{Q}{4\pi D \rho c t_{\max}} \left(1 - \exp\left(-\frac{t_p}{t_{\max}}\right)\right) \quad \text{Eq. 6,}$$

where Q is the applied heat per unit length (J m^{-1}), D the tangential thermal diffusivity ($\text{m}^2 \text{s}^{-1}$), and t_p the heating duration (s). Thermal conductivity (k) was estimated using the classical heat transfer relationship between D and ρc . Solving for ρc gives Equation 7.

$$\rho c = \frac{Q \left(1 - \exp\left(-\frac{t_p}{t_{\max}}\right)\right)}{4\pi D t_{\max} \Delta T} \quad \text{Eq. 7.}$$

RESULTS AND DISCUSSION

Estimation of sap flow velocity

Estimated V_h from the VPD-driven advection–diffusion model ranged between 3 and 5 cm h^{-1} (Table 1). The similar maximum, minimum, and mean values of T_{sap} , $\Delta T_{\max}$ after 62 s, and V_h across the three sensors indicate that all SFS sensors exhibited comparable performance during the measurement period. These V_h values correspond to relatively low absolute SF rates ($0.02\text{--}0.03 \text{ kg h}^{-1}$) due to the small trunk cross-sectional area ($\approx 7 \text{ cm}^2$) and to the decreased atmospheric driving for transpiration in a greenhouse. When normalized for larger basal areas in field-grown olive trees ($\approx 165\text{--}197 \text{ cm}^2$; Tognetti et al., 2004), the estimated V_h would correspond to absolute $\text{SF} \sim 0.5\text{--}1 \text{ kg h}^{-1}$, consistent with typical literature values of $2\text{--}12 \text{ cm h}^{-1}$. Similar magnitudes have been reported for young or container-grown plants (Lopez-Bernal et al., 2010; López-Bernal et al., 2017).

Table 1. Sap flow sensor measurements (SFS) (17 Oct–2 Nov): maximum, minimum, and mean of pre-heating sap temperature (T_{sap} , °C), maximum temperature difference at 62 s (ΔT_{max} (62), °C), sap flow velocity (V_h , cm h⁻¹), and vapor pressure deficit (VPD, kPa) for SFS sensors 1–3.

	T_{sap} (°C)			ΔT_{max} at 62s (°C)			V_h (cm h ⁻¹)			VPD (kPa)		
	SFS 1	SFS 2	SFS 3	SFS 1	SFS 2	SFS 3	SFS 1	SFS 2	SFS 3	SFS 1	SFS 2	SFS 3
Max	33.72	34.16	35.15	5.60	4.66	5.05	4.83	5.01	5.00	3.95	3.53	3.61
Min	22.97	23.06	23.02	2.98	3.13	3.38	0.00	0.00	0.00	0.00	0.11	0.18
Mean	26.85	27.16	27.12	4.53	3.69	4.24	0.54	0.69	0.59	0.55	0.65	0.65

Revisiting heat-pulse interpretation under low-flow, small-stem conditions

In small-diameter stems (~3 cm) with low SF, ΔT —the temperature rise 60 s post-heating pulse versus pre-heating baseline—was higher daytime than night-time, opposing classical advective views where advection boosts downstream heat transport and raises ΔT . This stems from the study's thermal regime and sensor setup: under low flow, Peclet number ($Pe = vL/\alpha$; v = sap velocity, L = heater-sensor distance, α = wood thermal diffusivity) signals diffusion-dominated transport (Figure 1). Heat spreads via conduction, confined by small stem volume and low radial xylem diffusivity; SFS configuration (thermistor at needle tip) amplifies this, with tangential diffusion along the needle accumulating heat near the sensor for larger ΔT even at minimal flow. Daytime amplification arises from lower tissue water, reduced k , and decreased ρc at warmer stem temperatures, yielding higher ΔT_{max} . Night-time cooler, hydrated xylem promotes radial dissipation, lowering ΔT . This aligns with advection-diffusion models as a diffusive-confinement limit: high flow lets advection displace thermal plumes (Marshall, 1958; Vandegehuchte and Steppe, 2012b), while low flow boosts ΔT_{max} via diffusion restrictions. The SFS configuration provides advantages in detecting subtle local heat dynamics: it allows absolute temperature measurements, improves sensitivity, and reduces alignment. Limitations include potential ΔT_{max} overestimation in small stems, strong dependence on wood thermal properties (k , ρc), and sensor-tissue contact, requiring careful calibration of the empirical coefficient Λ .

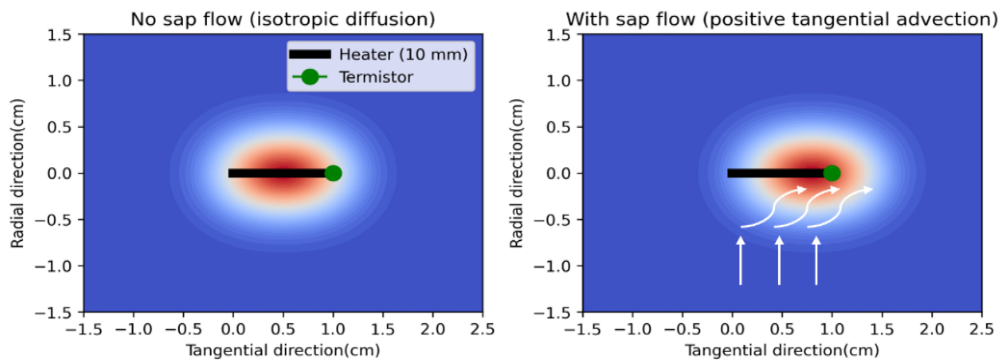


Figure 1. Tangential thermal propagation in olive tree (10 mm heater, thermistor at the tip).

Relationship between the thermal parameters and vapor pressure deficit

The thermal parameters estimated from the VPD-based sap velocity model are consistent with those reported in literature (Table 2). (Chen et al., 2012) reported D ca. $2.2 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, while (Vandegehuchte and Steppe, 2012a) observed values from 2.2 to $2.71 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$. The estimated ρc aligns well with estimates for wet wood and sapwood, matching materials with substantial water content (Forster, 2019; Vandegehuchte and Steppe, 2012b). The derived thermal conductivity matches the upper range reported for saturated

or near-saturated wood ($0.6\text{--}0.8\text{ W m}^{-1}\text{ K}$). This reinforces the physical realism of the parameters, indicating that the model produced thermophysical coefficients compatible with sapwood materials. Overall, the close correspondence between the modelled parameters and published ranges (Chen et al., 2012; Forster, 2019; Vandegehuchte and Steppe, 2012a, 2012b) supports the validity of using VPD-derived velocities to infer thermal properties in situ. The agreement between the measured mean values and the literature-based expected ranges supports the physical consistency of the velocity model output.

Table 2. Summary statistics of thermal parameters: maximum, minimum, and mean values of thermal diffusivity ($D1\text{--}3$, $\text{m}^2\text{ s}^{-1}$), volumetric heat capacity ($\rho c1\text{--}3$, $\text{J m}^{-3}\text{ K}^{-1}$), and thermal conductivity ($k1\text{--}3$, $\text{W m}^{-1}\text{ K}^{-1}$) for hydrated sapwood.

	<i>D1</i>	<i>D2</i>	<i>D3</i>	<i>ρc</i>	<i>ρc</i>	<i>ρc</i>	<i>k1</i>	<i>k2</i>	<i>k3</i>
max	8.34E-07	2.25E-06	8.55E-07	4.69E+08	6.69E+07	4.50E+07	1.14E+00	1.09E+00	1.01E+00
min	1.77E-09	1.38E-08	1.92E-08	8.95E+05	4.30E+05	8.83E+05	6.09E-01	7.32E-01	6.75E-01
mean	1.10E-07	1.15E-07	1.15E-07	1.52E+07	1.32E+07	1.06E+07	7.57E-01	9.28E-01	8.09E-01

Expected variation in max/min values arises from spatial heterogeneity and measurement noise, but consistent parameter estimates affirm the physical grounding of sap velocity inferred from ΔT (or $1/\Delta T$). Soil moisture held steady (Table 3), indicating well-watered plants. Given soil water conditions, sap flow variances stem mainly from atmospheric drivers, not soil water availability. Figure 2 depicts ρc -VPD relationships for October 26 and 29, alongside soil moisture. Both days reveal a negative ρc trend with rising VPD, tying higher evaporative demand to diminished sapwood thermal storage. October 29 showed stable soil moisture, a flatter ρc -VPD response with lower maxima, reduced atmospheric demand, lower V_h , conservative T , and consistent sapwood hydration.

Table 3. Soil sensor measurements: maximum, minimum, and mean values of volumetric water content (H , %), soil temperature (T , $^{\circ}\text{C}$), and soil electrical conductivity (CE , mS cm^{-1}) recorded by soil sensors Sofis1–3.

	<i>H</i> (%)			<i>T</i> ($^{\circ}\text{C}$)			<i>CE</i> (mS/cm)		
	sofis1	sofis2	sofis3	sofis1	sofis2	sofis3	sofis1	sofis2	sofis3
Max	100.00	50.00	70.00	25.80	26.30	26.30	359.00	681.00	546.00
Min	97.00	46.00	58.00	18.80	18.50	17.80	284.00	612.00	502.00
Mean	99.67	47.34	59.48	22.03	22.49	21.82	333.22	649.25	521.85

Calibration of sap flow sensors

Daily plant T obtained with lysimeters, and SF sensors is presented in Figure 3a. Actual mean daily transpiration was close to 2 and 0.5 L/d per plant for the periods of July and October, respectively. Around 50% of the variance of transpiration was explained by daytime vapor pressure deficit during the summer measurements. Comparison of daily plant T is presented in Figure 3. The regression analysis for the whole data set indicates a good correlation (ca. $R^2=0.9$) between daily T values obtained with the precision weighing lysimeters and the ones obtained with SF. There is an underestimation by SFS below 10%, for daily values of T , which is an interesting result, although further data retrieval is necessary for confirmation.

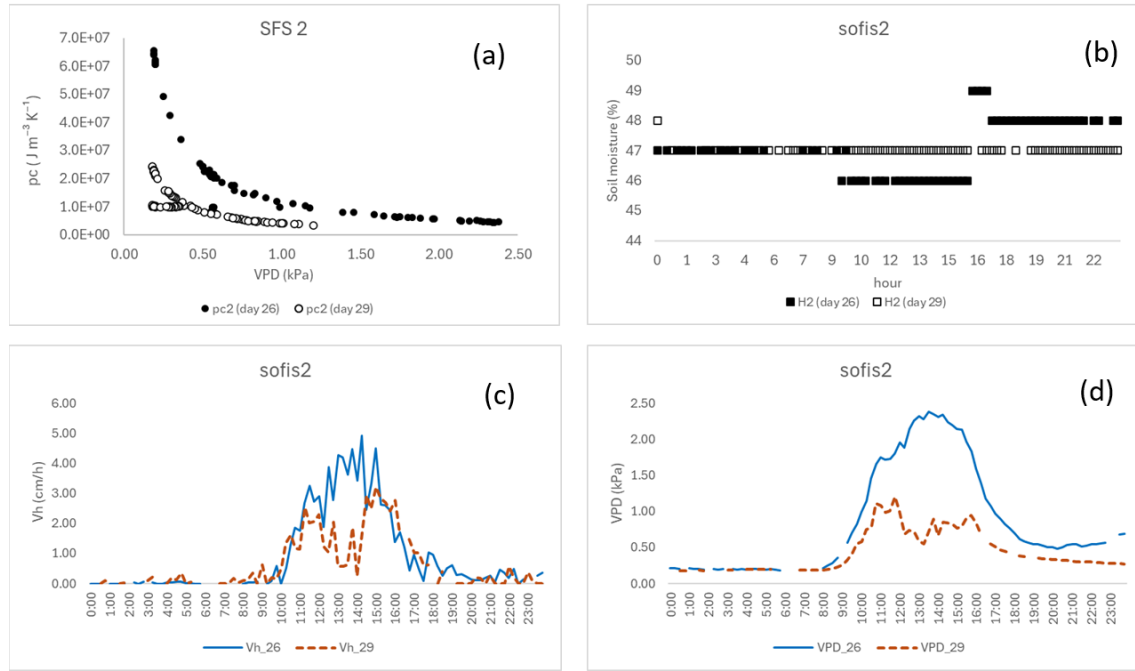


Figure 2. Daily patterns of soil moisture, sap flow velocity (V_h), and vapor pressure deficit (VPD) on October 26 (DOY 299) and 29 (DOY 302). a) volumetric heat capacity (ρc) vs. VPD; b) soil moisture; c) V_h October 26; d) VPD October 29.

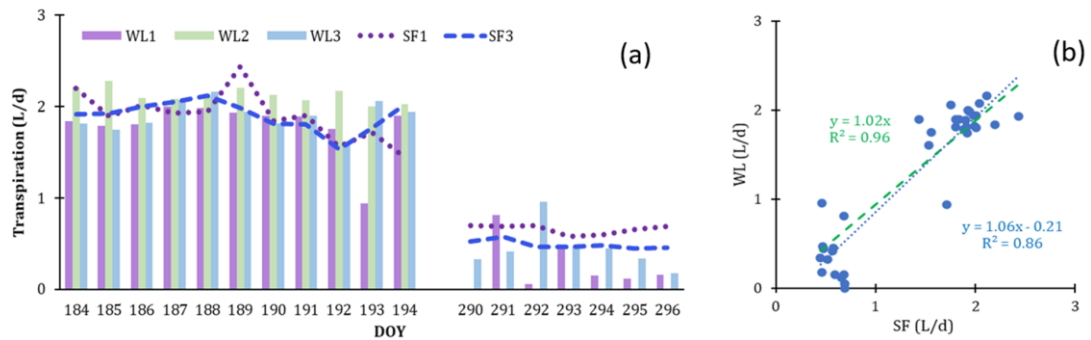


Figure 3. Daily transpiration measured by precision weighing lysimeters (WL) and sap flow (SF) sensors during July (184-194) and October (290-296) 2025; a) comparison; b) regression analysis; dashed green line - regressed to origin; dashed blue line - not regressed to origin.

CONCLUSIONS

The VPD-based advection-diffusion model implemented in the SOFIS system could reliably estimate sap flow velocity under controlled conditions. The low cost and simple design of the sensor makes it suitable for wider deployment, enabling continuous monitoring at multiple points. The sensitivity of the thermal signal to VPD also indicates that SOFIS can detect early changes in transpiration, helping to prevent water stress. In agricultural practice, the system would compare sap flow readings against an empirical baseline under the same VPD and significant deviation would trigger the irrigation system. Sap flow sensors calibration showed an underestimation under 10% and a high correlation to precision weighing lysimeters data. Further field validation is needed to confirm performance under variable conditions but SOFIS offers a practical and scalable approach to improve irrigation efficiency.

ACKNOWLEDGEMENTS

This work was supported by FCT – Fundação para a Ciência e Tecnologia, I.P. through 1) project UID/04129/2025 (<https://doi.org/10.54499/UID/04129/2025>) of LEAF-Linking Landscape, Environment, Agriculture and Food and 2) project 2024.14195.MIT - Adapting Precision Irrigation to Climate Change using low-cost sensors and information technology (ApiCC) with national funds.

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